

Should we use $(rI - A)$ or $(A - rI)$ to set up the characteristic polynomial of A , and the homogeneous system to find eigenspace?

Answer:

The short answer is that both methods are correct and will lead you to the same final answers, but one convention is often preferred for its algebraic properties.

Let's break down why both work and which one you might want to choose.

The Characteristic Equation (Finding Eigenvalues)

The definition of an eigenvalue r and its corresponding eigenvector $\mathbf{v}$ for a matrix $\mathbf{A}$ is:

$$\mathbf{A}\mathbf{v} = r\mathbf{v}$$

where $\mathbf{v}$ is a non-zero vector. To solve this, we rearrange the equation to form a homogeneous system.

$$\mathbf{A}\mathbf{v} - r\mathbf{v} = \mathbf{0}$$

Here is where the two paths diverge:

Path 1: Using $(\mathbf{A} - rI)$

This is the more direct rearrangement. We can factor out the vector $\mathbf{v}$ to get:

$$(\mathbf{A} - rI)\mathbf{v} = \mathbf{0}$$

For this equation to have a non-zero solution for $\mathbf{v}$, the coefficient matrix $(\mathbf{A} - rI)$ must be singular, which means its determinant must be zero. This gives us the characteristic equation:

$$\det(\mathbf{A} - rI) = 0$$

Path 2: Using $(rI - \mathbf{A})$

Alternatively, we could have rearranged the original equation $\mathbf{A}\mathbf{v} = r\mathbf{v}$ differently:

$$r\mathbf{v} - \mathbf{A}\mathbf{v} = \mathbf{0}$$

Factoring out $\mathbf{v}$ gives:

$$(rI - \mathbf{A})\mathbf{v} = \mathbf{0}$$

Similarly, for this to have a non-zero solution for $\mathbf{v}$, the coefficient matrix $(rI - \mathbf{A})$ must be singular. This gives the characteristic equation:

$$\det(rI - \mathbf{A}) = 0$$

Why do they give the same eigenvalues?

The two matrices are negatives of each other: $(rI - \mathbf{A}) = -1 \cdot (\mathbf{A} - rI)$. A property of determinants states that for an $n \times n$ matrix $\mathbf{B}$, $\det(k\mathbf{B}) = k^n \cdot \det(\mathbf{B})$. Therefore:

$$\det(rI - \mathbf{A}) = \det(-1 \cdot (\mathbf{A} - rI)) = (-1)^n \det(\mathbf{A} - rI)$$

Since we are setting the determinant to zero, the $(-1)^n$ factor doesn't change the roots of the polynomial. $(-1)^n \det(\mathbf{A} - rI) = 0$ is true if and only if $\det(\mathbf{A} - rI) = 0$.

The resulting characteristic polynomials will either be identical (if n is even) or negatives of each other (if n is odd), but they will always have the same roots, which are the eigenvalues.

- **$\det(rI - \mathbf{A})$:** This form is often preferred in more abstract or theoretical mathematics. The reason is that the resulting characteristic polynomial $p(r) = \det(rI - \mathbf{A})$ is **monic**, meaning its leading coefficient (the coefficient of r^n) is $+1$. Monic polynomials have some nice algebraic properties and are considered "cleaner."

- **$\det(\mathbf{A} - r\mathbf{I})$:** This form is very common in introductory textbooks and applied fields. It arises more directly from the standard rearrangement $\mathbf{A}\mathbf{v} - r\mathbf{v} = \mathbf{0}$. The leading coefficient of the polynomial will be $(-1)^n$.

The Homogeneous System (Finding Eigenspaces)

Once you find an eigenvalue r , you must solve for the corresponding eigenvectors by finding the solution space of the homogeneous system you used:

$$(\mathbf{A} - r\mathbf{I})\mathbf{v} = \mathbf{0} \text{ or } (r\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}.$$

These two systems are equivalent. One is just the negative of the other. Solving $\mathbf{B}\mathbf{x} = \mathbf{0}$ gives the exact same set of solutions as solving $-\mathbf{B}\mathbf{x} = \mathbf{0}$. Therefore, you will find the exact same eigenspace (the set of all eigenvectors for a given eigenvalue, plus the zero vector) regardless of which form you use.

Conclusion:

You can use either $(r\mathbf{I} - \mathbf{A})$ or $(\mathbf{A} - r\mathbf{I})$ to set up both the characteristic polynomial and the homogeneous system. Both will yield the correct eigenvalues and eigenspaces.

The convention used in the course MA1513 is $(r\mathbf{I} - \mathbf{A})$.